

Basic

1. (a) Divide $2x^3 + 5x^2 + 6$ by $x^3 + 2x$. [1]
 (b) Express $\frac{2x^3 + 5x^2 + 6}{x^3 + 2x}$ in partial fractions. [5]
 [N22/I/3]
2. Express $\frac{13x - 6}{x^2(2x - 3)}$ in partial fractions. [6]
 [N21/I/5]
3. Show that $x = -1$ is a solution of the equation $3x^3 + 4x^2 - x - 2 = 0$ and hence solve the equation completely. [5]
 [N21/II/2]
4. Express $\frac{4x^2 - 7x + 9}{2x^2 - x - 3}$ in partial fractions. [5]
 [N20/I/3]
5. (i) Show that $x - 1$ is a factor of $x^3 + x^2 - x - 1$. [1]
 (ii) Express $\frac{4}{x^3 + x^2 - x - 1}$ as the sum of three partial fractions. [7]
 [N19/II/3]
6. Express $\frac{7x^2 - 12x + 17}{(2x - 1)(x^2 + 4)}$ in partial fractions. [6]
 [N18/I/3]
7. (i) By using long division, divide $3x^3 - x^2 + 27x - 9$ by $3x - 1$. [1]
 (ii) Express $\frac{6 + 11x - 5x^2}{3x^3 - x^2 + 27x - 9}$ in partial fractions. [5]
 [N17/I/8]
8. Express $\frac{2x^2 + 4x - 31}{x^2 + x - 6}$ in partial fractions. [5]
 [N16/I/5]

Intermediate

9. It is given that $f(x) = x^3 + ax^2 - 3x + b$, where a and b are constants, has a factor of $x + 2$ and leaves a remainder of 30 when divided by $x - 3$.
 (a) Find the values of a and b . [4]
 (b) Using these values of a and b , factorise $f(x)$ completely into three linear factors, using surds where necessary. [3]
 [N22/II/2]
10. A polynomial, P , is $2x^3 - x^2 - 13x + k$, where k is a constant.
 (a) Find the value of k given that P leaves a remainder of 6 when divided by $x - 2$. [2]
 (b) In the case where $k = -6$, the quadratic expression $2x^2 + ax - 3$ is a factor of P . Find the value of the constant a . [4]
 [N21/I/6]

11. (a) The remainder when $x^3 + ax$, where a is a constant, is divided by $x - 2$ is the same as the remainder when it is divided by $x + 1$. Find the value of a . [3]
- (b) Solve the equation $x^3 - 2x^2 - 4x + 3 = 0$, expressing non-integer roots in surd form. [4]
- [N20/I/8]
12. The equation of a polynomial is given by $p(x) = 2x^3 + 5x^2 - 18$.
- (i) Find the remainder when $p(x)$ is divided by $x + 2$. [2]
- (ii) Show that $2x - 3$ is a factor of $p(x)$. [2]
- (iii) Show that the equation $p(x) = 0$ has only one real root. [4]
- (iv) Use your answers to parts (ii) and (iii) to solve the equation $2^{3y+1} + 5(2^{2y}) = 18$. [3]
- [N18/II/8]
13. (i) Find the remainder when $2x^3 - 3x^2 - 5$ is divided by $2x + 1$. [2]
- (ii) Factorise completely the cubic polynomial $2x^3 - 3x^2 + 1$. [4]
- (iii) Express $\frac{4 - 5x - 8x^2}{2x^3 - 3x^2 + 1}$ as the sum of 3 partial fractions. [4]
- [N15/II/8]